

# Supplementary data

Guidance and surveillance capability for bystander *Staphylococcus aureus* tetracycline resistance under doxy-PEP: a three-stream evidence audit

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All supplementary material is relocated from the validated analysis and regenerates from the public repository (`make all; outputs/`), archived with a citable DOI. *JAC* nomenclature conventions are followed and MICs are expressed in mg/L.

## S1. The measurement-inheritance schema

We make the construct operational so it can be applied beyond this case. Treat the answerability of a question as requiring a measurement vector

$$M = \{U, E, P, D, T, L, R\}$$

where  $U$  is the unit/population,  $E$  the exposure,  $P$  the phenotype/outcome,  $D$  the denominator,  $T$  the temporal structure/cadence,  $L$  the linkage at which  $E$  and  $P$  are joined, and  $R$  the locus of measurement responsibility. Measurement inheritance occurs when a downstream system retains the dimensions appropriate to the upstream question but not those the downstream estimand requires. A distributive corollary follows when the benefit and the potential downstream harm accrue to different populations ( $B \neq H$ ): the beneficiary population  $B$  remains represented by the inherited  $\{U, E, P, D, L\}$ , while the population  $H$  that would bear an externalised consequence is represented by none of them jointly. The three streams instantiate the schema directly (Supplementary Table 1).

The two critical gaps are  $R$  in Stream B (no party assigned to measure the bystander) and  $L$  in Stream C (no deployed system joins doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator).

## S2. Stream A — supplementary trial analyses

These analyses establish that the cross-sectional carriage endpoint could not resolve the downstream population question under the observed designs and sampling structure. They are MRSA-scoped (MRSA being a subset, not the analytic unit) and do *not* bear on the individual-level result the final trial resolved.

- **Detectability (interim).** For each first-party primary-trial *S. aureus* arm comparison, exact-Fisher power and the minimum detectable RR at 80% power were computed from arm sizes and control rate against the external benchmark. None of four comparisons was powered; the minimum detectable RR was 3.8–5.1 against a benchmark of 1.42–2.25 (`outputs/detectability_result.md`).
- **Duration/accrual.** Preserving each comparison's arm ratio and control rate, the sample reaching 80% power lands in the hundreds to thousands per arm; the realised denominator

Supplementary Table 1: Measurement dimensions required to follow the doxy-PEP resistance signal across the evidence chain. The measurement vector comprises unit/population ( $U$ ), exposure ( $E$ ), phenotype/outcome ( $P$ ), denominator ( $D$ ), temporal structure ( $T$ ), linkage ( $L$ ), and measurement responsibility ( $R$ ). “Present” indicates that the dimension is available at the resolution required by the downstream question; “wrong unit/resolution” indicates that the dimension exists but is not aligned with the target estimand; “absent” indicates that it is not captured.

| Dimension          | Stream A (trial)             | Stream B (guidance)                | Stream C (surveillance)                      |
|--------------------|------------------------------|------------------------------------|----------------------------------------------|
| $U$ unit           | Participant <i>S. aureus</i> | Not specified for <i>S. aureus</i> | MRSA subset (wrong unit)                     |
| $E$ exposure       | Randomised doxy-PEP          | Prescribing context                | PrEP proxy only                              |
| $P$ phenotype      | Incident doxy-R (present)    | Not measured (absent)              | Methicillin axis (wrong unit)                |
| $D$ denominator    | Randomised participants      | None for <i>S. aureus</i>          | Population, but methicillin                  |
| $T$ cadence        | Time-to-event                | None                               | Cross-sectional/facility                     |
| $L$ linkage        | Individual                   | None assigned                      | No common $E$ – $P$ denominator (absent)     |
| $R$ responsibility | Trial investigators          | Unassigned (absent)                | Unassigned for tetracycline <i>S. aureus</i> |

reflects the early efficacy stop (`outputs/duration_result.md`). No duration counterfactual is inferred from the two arm-split visits.

- **Clustering dispersion.** Between-cohort dispersion of MRSA colonisation across independent MSM cohorts was estimated by a logit-normal binomial random-effects model (Gauss–Hermite quadrature) as a conservative external heterogeneity anchor (`outputs/dejong_sigma_result.md`).
- **Trend-test degradation.** At the trials’ per-visit denominators, a naive cross-sectional trend test (Cochran–Armitage) is badly oversized under the estimated dispersion, scoring a flat process as a significant trend most of the time (`outputs/coverage_null_result.md`).
- **Outbreak-shape non-identifiability.** Flat, monotone-trend and outbreak-wave models fitted to the MRSA carriage series by maximum likelihood are not distinguishable by AIC or parametric-bootstrap likelihood-ratio tests (`outputs/three_outbreak_fit_result.md`).
- **Mechanism-blind measurement in *S. aureus*.** In a military antimalarial–doxycycline cohort of 168 personnel, overall tetracycline-class resistance in *S. aureus* did not differ by doxycycline exposure while *tet*(M) genes were significantly more common among the exposed ( $P = 0.031$ ) — a single breakpoint concealing a mechanism shift. This cohort is *S. aureus*-specific (not cross-organism) (Mende *et al.*, *Diagn Microbiol Infect Dis* 2016;86:211–220). The two determinants also differ in mobility: *tet*(M) is carried on conjugative transposons that cross strain boundaries, whereas *tet*(K) is typically plasmid-borne (Roberts and Mullany, *FEMS Microbiol Rev* 2011;35:856–871; Han *et al.*, *J Clin Microbiol* 2007;45:1350–1352), so the single breakpoint that merges the two phenotypes also merges two different dissemination routes.

The one positive within-arm trend statistic in this material is a published-figure re-analysis of DOXYVAC MRSA carriage (1.8% to 9.9%), cited as such and not originated here.

### S3. Stream B — guidance corpus and coding

**Denominator.** The purposive denominator is the US jurisdictions with documented community-associated MRSA/USA300 transmission in men who have sex with men, defined by de Jong’s systematic review and the outbreak literature it cites — the same source supplying the Stream A clustering dispersion. Governmental public-health guidance only was assessed; clinic and provider-organisation documents were excluded because they carry neither surveillance authority nor monitoring policy.

**Coding.** Each document was coded from full text (not search snippets), one record per document, every finding carrying a locator, for: whether it mandates in-category monitoring (gonorrhoea, chlamydia, syphilis) and host-toxicity labs; and whether it names, counsels on, or measures *S. aureus*. Bystander treatment was graded silent / generic-microbiome / organism-named; *S. aureus* monitoring as required / suggested / silent / explicitly-none (the last requiring a verbatim quote). Coding was blind to the expected outcome. The author identified, retrieved and screened all source documents for eligibility from full text before any coding pass; a first LLM pass coded all 10 documents against the prespecified schema, and a second, blinded LLM pass (a chat interface rather than the command-line agent, same model family) independently re-coded a 5-document subsample (25 field-codings, chosen to span all *S. aureus*-location categories), with disagreements resolved by the author against the source. Per-field agreement is reported as raw concordance: 92% (23/25), and the single field with meaningful disagreement (bystander treatment;  $\kappa = 0.58$ , context only) is reported unrounded. Because the two passes share a model family their errors are not independent (main-text Limitations).

**Result.** Of the six documented-outbreak jurisdictions — San Francisco, Chicago, New York City, Boston (Massachusetts), Los Angeles County and San Diego — 0 require or suggest *S. aureus* monitoring; three name *S. aureus*, two invoke only a generic microbiome risk, one is silent. An independence check from document content (structure, shared passages, stated derivation) finds four independently authored declines, with Los Angeles County adapted from San Francisco and Massachusetts from an external template. Across the full coded corpus (10 documents: CDC, WHO and eight US jurisdictions) 0 instrument the bystander; a further jurisdiction deferring to CDC inherits CDC’s non-monitoring and is counted separately. The full per-document coding table with locators is in the repository ([data/processed/](#)).

### S4. Stream C — surveillance enumeration and dilution model

**Enumeration.** A prespecified set of 12 established US surveillance and data systems (federal, state or metropolitan) plausibly capable of capturing, at the required level, either doxy-PEP/doxycycline exposure or *S. aureus* AMR phenotype was coded on three capabilities: exposure capture, phenotype capture (of the *S. aureus* tetracycline unit, not the MRSA subset) and the join. Two capture exposure at a population denominator (AIDSVu; CDC AtlasPlus) as a PrEP proxy; zero capture *S. aureus* tetracycline resistance at a population denominator. The population-based *S. aureus* platform (ABCs/EIP) and the National Healthcare Safety Network report the methicillin axis; the facility cumulative antibiogram is the only routine tetracycline reading and carries no population denominator (CLSI M39 mandates the MRSA/MSSA split and makes tetracycline a selectively suppressible supplemental agent); the National Antimicrobial Resistance Monitoring System excludes *S. aureus*. Tetracycline-resistant *S. aureus* is not nationally notifiable in the United States. Surveillance capability was assessed as of August 2026; Stream C was coded by a single pass (unlike the double-coded Stream B), an asymmetry noted in the main-text Limitations. The full inclusion/exclusion list with a reason per system is in Supplementary Table 2.

**Dilution model.** The exposed doxy-PEP population is indexed on the combined PrEP and

Supplementary Table 2: US surveillance and data systems assessed for doxy-PEP exposure and *S. aureus* tetracycline phenotype. Twelve systems were included because they could plausibly capture either exposure or antimicrobial-resistance phenotype at the required level; three additional systems or data sources were excluded with reasons shown. None of the 12 included systems captures both doxy-PEP exposure and *S. aureus* tetracycline phenotype at a common population denominator.

| System                                              | Capability / reason                                                                   |
|-----------------------------------------------------|---------------------------------------------------------------------------------------|
| <i>Included (12)</i>                                |                                                                                       |
| AIDSVu                                              | Exposure only, PrEP proxy at population denominator; no phenotype                     |
| CDC NCHHSTP AtlasPlus                               | Exposure only, PrEP proxy at population denominator; no phenotype                     |
| NHSN Antimicrobial Use Option                       | Inpatient antimicrobial use, no population denominator; not doxy-PEP                  |
| State PDMPs                                         | Do not capture doxy-PEP dispensing as such; no phenotype                              |
| Metro STD/HIV sentinel surveillance                 | Sentinel exposure, no population denominator; no <i>S. aureus</i> phenotype           |
| NARMS                                               | Excludes <i>S. aureus</i> altogether                                                  |
| GISP / eGISP                                        | Gonococcal isolates only (wrong organism)                                             |
| ABCs / EIP invasive <i>S. aureus</i>                | Population-based, but methicillin (MRSA/invasive) — wrong phenotype unit              |
| NHSN AR Option / MRSA LabID                         | MRSA events; facility, no population denominator                                      |
| AR Lab Network (ARLN)                               | Isolate referral; no exposure link, no population denominator                         |
| NNDSS                                               | Tetracycline-resistant <i>S. aureus</i> is not notifiable                             |
| Hospital cumulative antibiograms                    | Only routine tetracycline reading, but non-standardised and no population denominator |
| <i>Excluded (3)</i>                                 |                                                                                       |
| CDC Find Doxy PEP directory                         | A provider/location directory, not surveillance                                       |
| Commercial AST networks (ATLAS, BD, PINC AI, IQVIA) | Commercial convenience datasets; no population denominator                            |
| Doxy-PEP RCTs and cohorts (DoxyPEP, DOXYVAC, Soge)  | One-off research studies, not deployed surveillance                                   |

PLWH male count, because the trial enrolled both (primary specification: every male person living with HIV counted as exposed, AIDSVu 2024 state PrEP and prevalence; male PLWH is a prevalence stock, and state prevalence carries no transmission-category split, which is why the whole male-PLWH count is used rather than a fractional MSM estimate). The exposed-subgroup dilution fraction is

$$f = \frac{(\text{male-PrEP} + \text{male-PLWH}) \text{ density}}{10^5} \text{ uptake } d \kappa,$$

where  $d$  is the share of users taking  $>3$  doses/month (the subgroup carrying the benchmark effect);  $f$  is population-independent because it tracks exposure density, not headcount. The minimum detectable change in a proportion near baseline  $R_0$  (two-sided  $\alpha = 0.05$ , 80% power) inverts to  $RR_{\text{needed}} = 1 + \text{MDE}/(fR_0)$ . Parameters are not uniformly generous: isolate volume  $N \in \{1\,000, 10\,000, 100\,000\}$  per geography-year is set generously high (favouring detection; 100 000 exceeds any real US stream), while realistic under-sampling ( $\kappa = 0.5$ ; grid  $\kappa \in \{0.2, 0.5, 1, 3, 5\}$ ) and the dose factor ( $d = 0.5$ ) bound against it; uptake  $\in \{20, 35, 55\}\%$  and  $R_0 \in \{5, 10, 13\}\%$ . Under the combined-denominator primary, the realistic cell requires  $RR_{\text{needed}} \approx 6\text{--}28$  across the design-effect range — far above any plausible effect. Detection appears in only one cell of the entire grid, where every generous assumption is maxed at once (55% uptake,  $5\times$  enrichment, 13% baseline, 100 000 isolates/geography-year and optimistic panel independence): there  $RR_{\text{needed}}$  falls to 1.03; under single-comparison inference even the most favourable cell needs  $RR_{\text{needed}}$  1.79. Against the more conservative cross-organism benchmark ( $RR$  1.42), the best metropolitan cell needs about twice ( $\approx 2\times$ ) the combined male exposure density of the densest US geography (Washington, D.C., 6 406 per 100 000 adult males in 2024), and realistic cells hundreds of times it. All values regenerate from `outputs/feasibility_plwh_result.md`, and Figures A–C show the combined PrEP+PLWH denominator (the primary specification); the more restrictive PrEP-only indexing gives  $RR \approx 14\text{--}64$ .

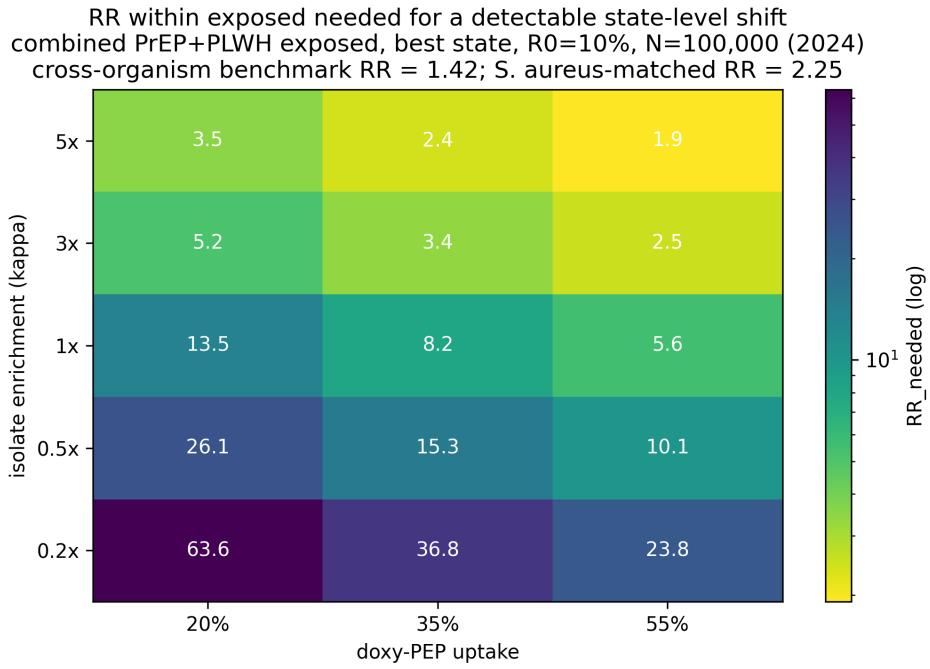
**Dose-threshold sensitivity.** The dose-above-threshold share  $d$  (the fraction of users taking  $>3$  doses/month, the subgroup carrying the benchmark effect) enters  $f$  linearly, so  $RR_{\text{needed}} - 1$  scales as  $1/d$ ;  $d = 1$  is the limiting case in which *every* user exceeds three doses/month. The realistic-cell verdict holds across the plausible range and degrades only at that bound, where the realistic lower bound (3.65) dips just below the trial’s hazard ratio (3.89) while still exceeding the cross-sectional benchmark band (Supplementary Table 3). Values regenerate via `dilution_plwh.dose_sensitivity`.

Supplementary Table 3: Sensitivity of the modelled surveillance threshold to the proportion of doxy-PEP users exceeding three doses per month.  $RR_{\text{needed}}$  is shown under the primary combined PrEP+PLWH exposure specification. The primary analysis assumes  $d = 0.50$ ;  $d = 1.00$  is the limiting case in which every doxy-PEP user exceeds three doses per month. DEFF, design effect.

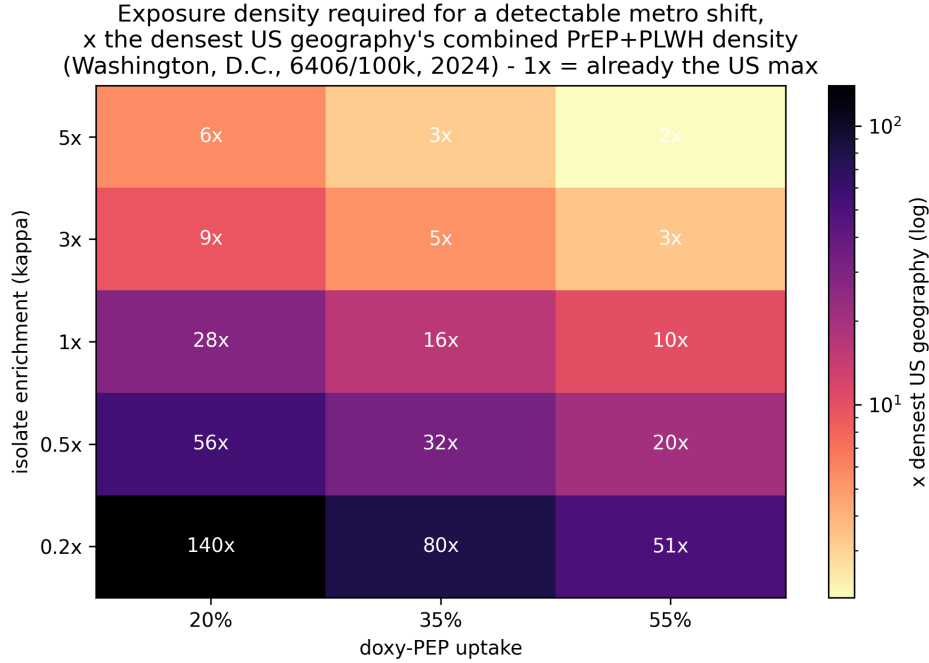
| $d$  | realistic (DEFF 1) | realistic (DEFF 25) | best cell (single-comp) | best cell (DEFF 1) |
|------|--------------------|---------------------|-------------------------|--------------------|
| 0.50 | 6.31               | 27.5                | 1.79                    | 1.03               |
| 0.75 | 4.54               | 18.7                | 1.52                    | 1.02               |
| 1.00 | 3.65               | 14.3                | 1.39                    | 1.01               |

Alt text: A heatmap grid of doxy-PEP uptake against isolate enrichment in which every cell shows a required relative risk above the cross-organism benchmark of 1.42.

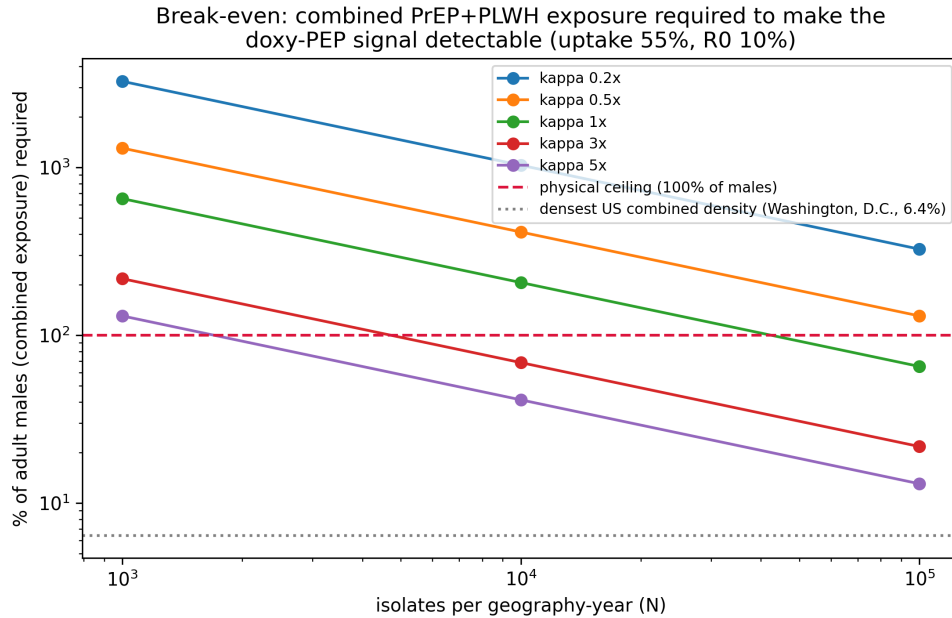
Alt text: A chart showing required combined PrEP+PLWH exposure density in multiples of the densest US metropolitan geography, with all requirements above the one-times national maximum.



**Figure A. State-level relative risk required for a detectable change in *S. aureus* tetracycline non-susceptibility.** Modeled  $RR_{needed}$  across doxy-PEP uptake and isolate-enrichment assumptions using the combined PrEP+PLWH exposure specification, the most favourable state exposure density,  $R_0 = 10\%$ , and 100 000 isolates per state-year. Every cell exceeds the conservative cross-organism benchmark of RR 1.42; the *S. aureus*-matched benchmark is RR 2.25.



**Figure B. Metropolitan exposure density required for detection of a population-level resistance shift.** Required combined PrEP+PLWH exposure density is expressed as a multiple of the densest observed US metropolitan geography, Washington, D.C. (6 406 per 100 000 adult males in 2024). Even under the most favourable modelled assumptions, detection requires approximately twice the highest observed exposure density.



**Figure C. Break-even exposure prevalence required for population-level detection.** Percentage of adult males who would need to be in the combined PrEP+PLWH exposed population for a detectable resistance shift, across isolate volume and enrichment assumptions. All modelled curves exceed the observed Washington, D.C. combined exposure density (6.4%); under realistic isolate volumes, required exposure prevalence also exceeds the 100% physical ceiling.

Alt text: Break-even curves of the share of all adult males under combined PrEP+PLWH exposure required for detection against isolate volume, all lying above the observed combined density (6.4%), and above the 100% physical ceiling under realistic isolate volumes.

## S5. Literature attention structure

A defined PubMed search (2015–2026; queries and a frozen result snapshot in the repository) identified 361 doxy-PEP papers, of which 13 (3.6%) mention a bystander staphylococcal organism and only 2 measure tetracycline-resistant *S. aureus* against a doxy-PEP exposure contrast. This is consistent with a research system whose attention remains concentrated where its existing instruments already point, and is not evidence of neglect by any author.

## S6. Distributive observability

The three streams share a measurement asymmetry that is a property of the evidence architecture before it is a moral claim. The intervention’s benefit and the trial’s resistance signal are identifiable within an exposed population: exposure ascertained, denominator defined, phenotype measured, a longitudinal causal contrast available. The population that would inherit a downstream resistance consequence has none of these jointly — no assigned measurement mandate (Stream B) and no deployed system in which doxy-PEP exposure and the *S. aureus* tetracycline phenotype share a denominator (Stream C). We call this difference in scientific visibility *distributive observability*: which population remains represented by the evidence system, and at what inferential resolution, once the question moves downstream from the treated to the affected population. It does not establish that a population harm has occurred; it establishes that, were one occurring, the deployed architecture could estimate the intervention side of the benefit–externality ledger far more completely than the externality side. An externality no instrument can price is not thereby small — it is unweighable, and an architecture that prices only one side of the ledger cannot show that benefit outweighs external cost. The construct is built from familiar parts (estimand mismatch; ascertainment and surveillance bias; metric substitution of the Goodhart type) but concerns their coupling across an evidence-production chain, and generalises to any intervention whose benefits concentrate in an observed recipient population while its consequences diffuse beyond it.

## S7. Reproducibility

All analysis is Python 3.11+ (validated on CPython 3.12.12; pandas, statsmodels, scipy, matplotlib); no proprietary dependencies. Raw data are immutable; all cleaning is in code; every figure and table regenerates from `make all`. Analysis decisions that could go either way were logged with dates in `DECISIONS.md` before results were known. All source files are SHA-256-pinned (`data/raw/**/CHECKSUMS.md`). The repository is public and archived with a citable DOI.